

VENKATA RAMA KRISHNA DULAM

AI & Systems Engineer · Security Researcher

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PROFESSIONAL SUMMARY

An innovative Computer Science & Engineering (AI) student at Amrita Vishwa Vidyapeetham. Specialist in developing intelligent agents, exploring quantum computing frameworks, and engineering low-level system architectures. Operates under the handle Cr@sh_R00t, focusing on the intersection of high-level AI models and high-performance system execution.

EDUCATION

Amrita Vishwa Vidyapeetham — Amritapuri Campus

B.Tech in Computer Science and Engineering (Artificial Intelligence)

2025 – 2029 (Expected)

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, C, JavaScript, Assembly, Jack
- **AI & Machine Learning:** Autonomous Agents (Negotiater AI, MediAgent), Neural Networks, NLP
- **Quantum Computing:** Qiskit, Gate-based Circuit Design, Quantum Algorithm Implementation
- **Low-Level Systems:** Compiler Design, Lexical Analysis, Hardware-Software Abstraction
- **Tools & Frameworks:** FastAPI, React, Tkinter, Git/GitHub, Linux (Kali linux/Ubuntu)
- **Security Research:** Mobile Binary Analysis, Reverse Engineering (Flutter/Dart)

PROJECTS

Negotiater AI

- Developed an intelligent agent designed for strategic communication and automated negotiation.
- Implemented decision-making logic to optimize outcomes in complex bargaining scenarios.
- **Technologies:** Python, AI Negotiation Frameworks

MediAgent

- Engineered a specialized AI agent tailored for the medical domain, leveraging high-yield biological and chemical data.
- Designed for intelligent query handling and information retrieval in medical contexts.
- **Technologies:** Python, NLP, Medical Knowledge Bases

Quantum Advantage Visualization (Qiskit)

- Implemented the Deutsch–Jozsa algorithm to demonstrate quantum computational speedup.
- Built a full-stack Python application with a Tkinter GUI for real-time visualization of quantum circuits and state probabilities.
- **Technologies:** Qiskit, Python, Tkinter

Jack Language Compiler Infrastructure

- Built a robust lexical analyzer and parser for the Jack programming language.

- Developed the tokenizer and parser architecture, integrated into an assembly-level repository for optimized execution.
- Performs syntax analysis and conversion of high-level code into optimized VM instructions.
- **Technologies:** Assembly, Python

Advanced Mobile Security Research

- Conducted offensive security research on Android internals, specializing in Flutter reverse engineering.
- Performed dynamic analysis and binary patching to understand application logic and data flow security.
- **Technologies:** Frida, Objection, Reflutter

ACHIEVEMENTS & INTERESTS

- **Quantum Computing:** Completed specialized training in quantum gate foundations and algorithms.
- **Systems Engineering:** Deep interest in the Nand2Tetris methodology — building a computer system from first principles.
- **Academic Focus:** Strong foundation in advanced Chemistry and Biology, complementing AI work in the medical-tech space.

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